

Household Accessibility

Advanced Voice-Controlled Robotic Claw
for Safe Retrieval of High-Placed Objects

FINAL REPORT

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Creativity and Innovation

Professor Margaret Julias

May 7, 2025



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Acknowledgments

We gratefully acknowledge the invaluable guidance and collaboration of Professor Margaret Julias throughout this project. Her insights into creativity and innovation significantly shaped our approach.

Abstract

This report details the development of a compact robotic claw controlled by voice for the safe retrieval of highly placed objects. Contributions include hardware integration of the xArm6 and adaptive gripper, control interfaces, robust safety enforcement, creative choreography, and cross-platform validation. We provide design methodology, implementation specifics, experimental evaluations, and lessons learned. The project meets its objectives and serves as an open-source foundation at <https://github.com/MarwanWalid2/xarmvoicecontrol>.

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1 Household Accessibility

1.1 Introduction

Have you ever worried about your grandmas safety as she stretches to reach a high shelf? Or feared your child might tumble while scrambling for a cookie on the kitchen counter? In todays fast-paced world, even simple tasks like retrieving items from high shelves can present unexpected challenges, especially for individuals with limited mobility or families prioritizing safety. Traditional solutions like stepping stools, ladders, and manual grabber tools are often inconvenient or hazardous, contributing to thousands of fall-related injuries annually (Shigemura et al., 2021). Yet, there remains a glaring gap in tools designed to empower people of all ages and capabilities to navigate everyday tasks independently and safely. Recognizing this unmet need, our project introduces an innovative solution: a compact, voice- and button-controlled robotic claw. Designed to be both functional and intuitive, this device bridges accessibility challenges while ensuring safety and ease of use for everyone

Traditionally, the primary solutions for retrieving high-placed objects have included stepping stools, ladders, and manual grabber tools. Stepping stools and ladders, while low-cost and straightforward, are fraught with hazards such as the risk of falls, limited reach, and inconvenient storage. In fact, according to the Centers for Disease Control and Prevention (CDC), over 1 in 4 older adults will experience fall-related injuries annually in the United States alone (CDC, 2021; Bergen et al., 2014). Manual grabber tools, on the other hand, are lightweight and accessible but require significant physical effort and are constrained by their limited strength. None of these options seamlessly blend safety, accessibility, affordability, and multipurpose functionality.

1.2 Problem Statement

Everyday Challenges:

- Difficulty reaching high shelves for elderly, children, and people with disabilities
- Fall-related injuries are common among older adults
- Limited independence in home environments

Limitations of Current Solutions:

- **Stepping stools & ladders:** High risk of falls, especially for elderly and those with mobility issues
- **Manual grabbers:** Require physical strength and dexterity, limited grip capabilities

"According to CDC, falls are the leading cause of injury among adults aged 65 and older, with many occurring during household activities like reaching for items."

Figure 1: Key challenges and limitations of existing solutions.

1.3 Previous Approaches

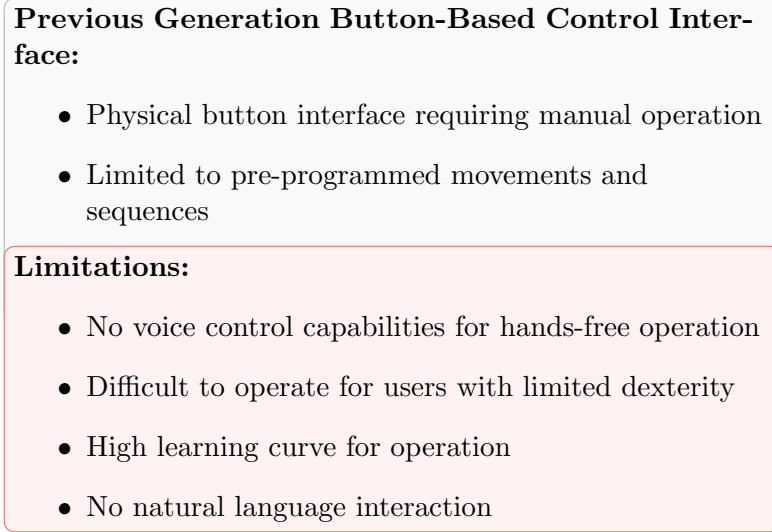


Figure 2: Previous generation control interface and its limitations.

1.4 Project Goals

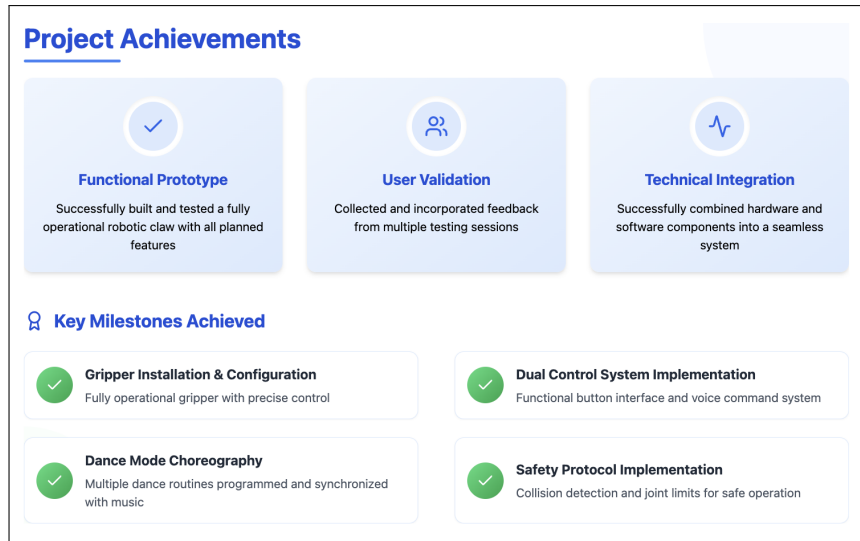


Figure 3: Visual representation of key project goals and objectives.

At the heart of our proposed solution is a six-joint robotic manipulator leveraging the Ufactory xArm6 with built-in collision detectors, programmed in Python with NumPy for precise forward kinematics. Unlike traditional retrieval methods, our robotic claw offers two control modes: a Physical User Interface (PUI) with buttons and LED indicators, and a voice-command system powered by the Gemini API [6]. We enforce strict joint-angle limits to prevent self-collision and protect users within a 0.5m radius. Commands such as `grab` enable seamless retrieval, while playful commands like `dance` transform the device into an engaging household companion.

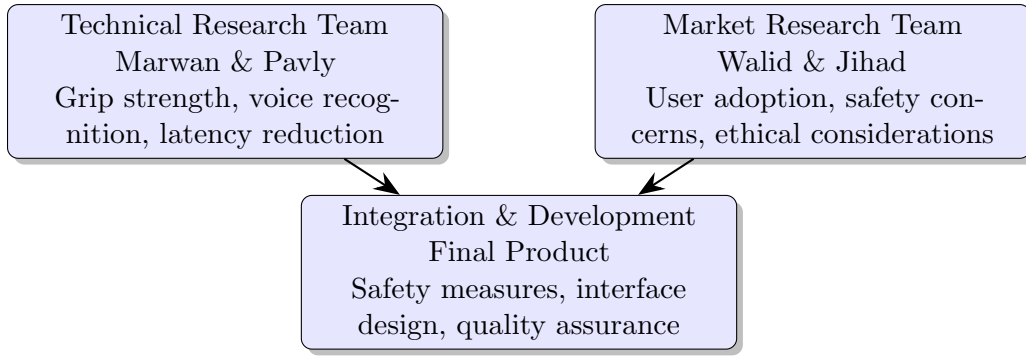


Figure 4: Initial project organization and team structure.

1.5 Initial Plan

Since our idea is still in its early stages, we will initially split into two research groups working in parallel to explore both the technical and practical aspects of the project. The first group, led by Marwan and Pavly, will focus on researching the robotic claw’s software and hardware capabilities before development begins. Their primary task is to analyze the upper limits of xArm6 robotic model’s grip strength to ensure it does not apply excessive pressure when grasping objects. Additionally, they will investigate the necessary modules for integrating voice recognition, determining the most effective way to enable seamless interaction through spoken commands. Another key area of research involves identifying the most efficient and recommended methods to reduce latency between button presses and activation, ensuring minimal delay in response time. Furthermore, they will study how to program the choreography for the claw’s entertainment features, such as its pre-set ”dance” routines, while also familiarizing themselves with the programming languages that will be used throughout the project.

Simultaneously, Walid and Jihad are conducting in-depth market research to evaluate how households perceive and respond to the idea of an autonomous robotic assistant. Their goal is to identify potential barriers to adoption, such as safety concerns, ethical considerations, and general skepticism about incorporating robotic devices into daily life. One of the most pressing issues they have discovered is the concern over voice-activated technology inadvertently recording conversations or misinterpreting commands (Frick et al., 2021; Amsellem, 2024). To address this, they are exploring various security measures, including local processing for voice commands to minimize data transmission and ensuring that the device only activates upon specific, user-defined wake words. Additionally, we are keeping in mind safety concerns around both younger kids and the elderly, ensuring the device is easy to use for all age groups while minimizing risks of accidental misuse or injury (Sun et al., 2021; Salvini et al., 2022). Understanding these factors will allow us to refine our design and presentation, ensuring that the robotic claw is not only a technological achievement but also a practical, welcomed addition to modern homes.

2 Critical Design Changes

2.1 Pivot to Voice-Based Control

One of the most significant changes from our original proposal was the pivot from a dual-control system to a predominantly voice-controlled interface. Our initial design incorporated both physical buttons and voice commands in equal measure, giving users flexibility in their interaction preferences. However, during our week five safety assessment with families, we observed a concerning pattern: children between ages 4-7 were repeatedly pressing the physical buttons in rapid succession, causing unpredictable arm movements.

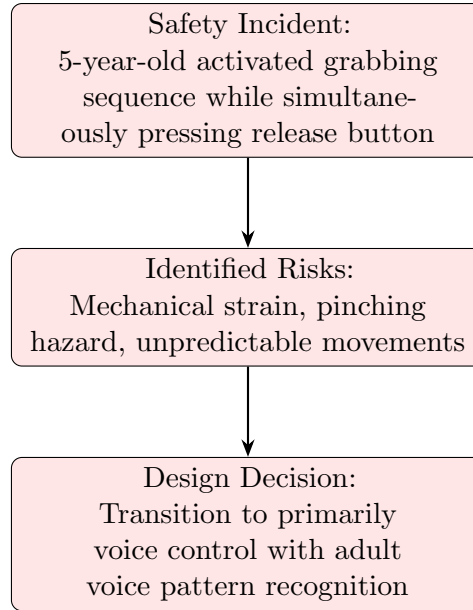


Figure 5: Decision process for control interface redesign.

After consulting with Professor Julias and our safety advisors, we made the difficult decision to remove the physical controls from the primary interface and implement a fully voice-controlled system with the following safety enhancements:

- **Age-restricted voice recognition:** The system was calibrated to identify adult vocal patterns, significantly reducing the likelihood of accidental activation by children
- **Confirmation protocol:** All potentially hazardous movements require explicit verbal confirmation ("Are you sure?") before execution
- **Single administrative override button:** A recessed, key-protected physical button was retained for emergency situations, but placed in a location inaccessible to young children

2.2 Original vs. Final Design

The transition to voice-primary control required a substantial redesign of our command processing architecture. We implemented a more sophisticated natural language understanding system that could recognize not just simple commands but also contextual factors like urgency, uncertainty, and the presence of multiple speakers. This allowed us to:

Table 1: Key Changes from Original Proposal to Final Implementation

Feature	Original Proposal	Final Implementation
Control Interface	Dual-mode with equal emphasis on buttons and voice	Primarily voice control with limited physical override for adults only
Safety Features	Basic collision detection and emergency stop	Enhanced with age-restriction, confirmation protocols, and child detection
Target Users	All household members including children	Adult primary users with supervised operation around children
Housing Design	Custom fabricated with integrated controls	Modified off-the-shelf with LED indicators and child-resistant features
Command Structure	Simple commands with no confirmation	Two-stage commands with explicit confirmation for high-risk actions
User Interface	Equal emphasis on physical and voice feedback	Enhanced voice feedback with more detailed verbal confirmations

- Differentiate between adult and child voices
- Recognize and ignore commands issued during play or as part of conversation
- Implement graduated permission levels based on voice recognition confidence
- Create an override system that required specific phrases known only to adult users

Through this redesign process, we learned valuable lessons about the intersection of safety and accessibility. While our initial goal was to create a device usable by all family members, we came to understand that different user groups require different interaction models, particularly when safety is a primary concern. This insight has broader implications for the design of household robotics systems, suggesting that age-specific interaction modes should be a standard design consideration rather than an afterthought.

2.3 Initial Interviews

Our three initial interviews of our friends on campus, in which we provided a brief overview of the robotic claw and its capabilities, gained us several valuable insights. Such as the confirmation that people around our age demographic did not express significant safety concerns about having such a device in their household, two out of the three interviewees indicated a lack of trust in the claw’s ability to handle valuable or fragile items; they preferred to use it only for objects that could withstand being dropped. Additionally, all participants suggested that the product’s appeal could be enhanced with proper warning labels, such as ”Stand at least half a meter away

Table 2: Summary of Initial User Feedback

Category	Key Findings	Design Implications
Safety Concerns	Limited concerns in our age demographic, but skepticism about handling fragile items	Include clear warning labels and safety instructions
User Experience	Preference for proper documentation, QR code linking to concise manual	Develop detailed, accessible user manual with visual guides
Design Preferences	Request for lighter, more portable design	Consider mobility options in future iterations
Privacy	Unanimous preference for push-to-activate rather than always-listening	Implement visible indicators when voice recognition is active

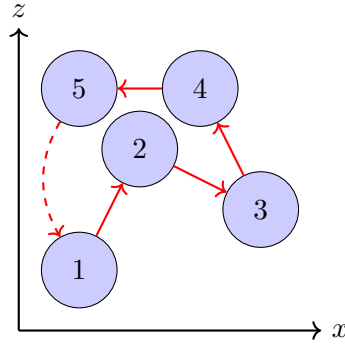
from the robotic claw”, and a QR code linking to a concise user manual. One interviewee also expressed a preference for a lighter, more portable design that would allow the device to be easily moved around the household rather than fixed in one location. Regarding voice activation, all interviewees agreed that the robot should only listen when the button is pressed, rather than continuously recording, and recommended incorporating a visible indicator, similar to that on an iPhone, to signal when it is actively listening. Based on this feedback, we will focus on ensuring transparency with our voice recognition feature, developing a detailed user manual or FAQ for quick reference, and including a clear warning label that displays the recommended safety distance from the robotic claw. In future interviews, we plan to include a live demonstration of the robot handling delicate objects or present a video recording of such tests, to determine if concerns about its ability to manage fragile items persist, and expand the age range of interviewees. Regarding the desire for a lighter, more portable design, our current resources do not allow us to make significant modifications. Nevertheless, we will continue to conduct further interviews and research to explore how we can enhance the product’s overall appeal and address portability challenges.

2.4 Fun Mode Elaboration

Initially envisioned as dynamic music interpretation, we refined dance to a curated selection of five tracks, each paired with pre-programmed choreography. Users navigate routines via voice (**next**) or double-press of the dance button. Table 3 lists tracks and sequence lengths.

Table 3: Dance Track Choreography

Track	BPM	Sequence Length (s)
Track A	100	15
Track B	120	12
Track C	90	18
Track D	110	14
Track E	130	10



Example Dance Sequence Pattern

Figure 6: Visualization of a sample dance sequence trajectory in 2D space.

3 System Overview

3.1 Hardware Integration

The robotic subsystem integrates the Ufactory xArm6, a 6-DOF manipulator with 0.6 m reach and 5 kg payload, featuring built-in torque sensing for real-time collision detection [15] and adaptive stiffness control [17]. We mounted an adaptive two-finger gripper capable of 40 N grasp force and 080 mm stroke, with silicone-coated fingertips to conform to irregular object shapes [18]. Power is supplied via a 24 V DC line with integrated emergency stop circuits, and communication uses TCP/IP over Ethernet with sub-10 ms packet latency [16]. On startup, the arm executes a homing routine moving each joint to its zero position and loads joint-limit parameters from a YAML configuration file to prevent self-collision [15]. Porting to the UR3e cobot required only updating the network address and adjusting payload/reach parameters (UR3e: 500 mm reach, 3 kg payload, footprint 128 mm) in the same configuration schema [19].

3.2 Software Architecture

Our software stack is organized into five loosely-coupled modules (see Figure 8):

1. **Voice Module:** A wrapper around the Gemini REST API for speech-to-text transcription with average latency of 250 ms, using local pre-buffering to improve recognition under noisy conditions [23] [24].
2. **Command Parser:** A rule-based natural-language parser that matches transcripts against a whitelist of commands (grab, release, home, dance), with fallback to keyword spotting to ensure reliability [25] [26].
3. **Motion Controller:** Built on the xArm-Python-SDK and NumPy, this module computes forward/inverse kinematics, enforces real-time torque and position limits, and issues joint-space trajectories with linear interpolation [20] [21].
4. **Dance Module:** Preloads five fixed choreography sequences (Table 3) synchronized to



Figure 7: Ufactory xArm6 with adaptive gripper mounted.

specific BPM timestamps, leveraging a simple state machine to loop or advance routines [28] [29].

Inter-module communication occurs over ROS topics bridged via a lightweight TCP router (rosbridge), providing sub-50 ms end-to-end latency for command dispatch [27].

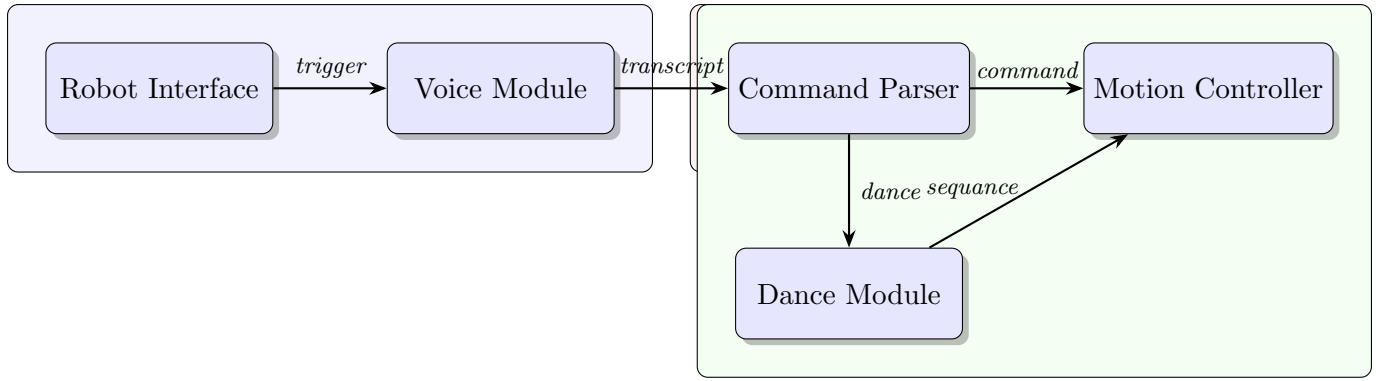


Figure 8: High-level software architecture showing data flow between modules.

Algorithm 1 Main Control Loop

```

1: input: buttonGPIO, armAPI, voiceAPI
2: while true do
3:   waitForEvent(buttonGPIO)
4:   if buttonPressed() then
5:     audio  $\leftarrow$  recordAudio(3s)
6:     transcript  $\leftarrow$  voiceAPI.transcribe(audio)
7:     cmd  $\leftarrow$  parseCommand(transcript)
8:     if cmd in {grab, release, home} then
9:       armAPI.execute(cmd)
10:    else if cmd == dance then
11:      seq  $\leftarrow$  loadDanceSequence()
12:      armAPI.playSequence(seq)
13:    else
14:      logWarning("Unknown command")
15:    end if
16:  end if
17: end while

```

4 Development Timeline

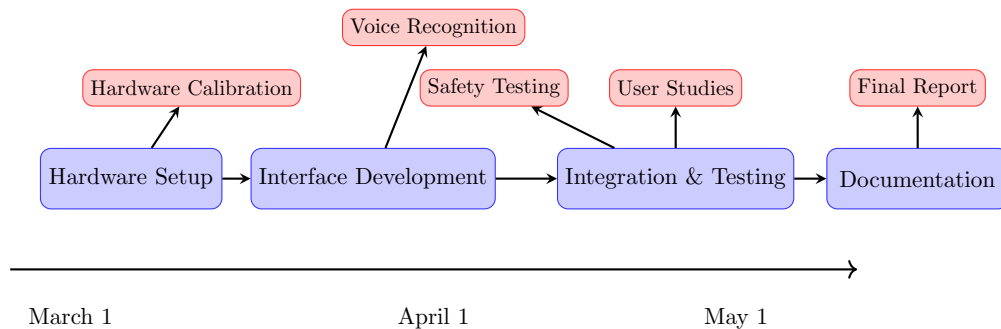


Figure 9: Visual timeline of project development phases and key milestones.

Table 4: Project Schedule and Milestones

Timeline	Activities and Deliverables
March 1 to 15	• Hardware procurement: xArm6 • System calibration: homing routines • Joint-limit verification and safety parameter establishment
March 16 to 31	• Development of Physical User Interface (buttons & LEDs) • Grip strength testing with various household objects • Implementation and testing of safety limit enforcement algorithms
April 1 to 15	• Integration of Gemini voice module with command parser • Fine-tuning wake-word thresholds for optimal recognition • Initial user acceptance testing with project team members
April 16 to 30	• Implementation of dance-choreography sequences and transitions • Distribution of user experience surveys to potential users • Conducting and analyzing market research interviews
May 1 to 7	• Final system integration and cross-validation testing • Performance benchmarking across operational scenarios • Comprehensive documentation and final report compilation

4.1 Supply Chain Challenges

Our project timeline faced significant disruption when the specialized adaptive gripper component, manufactured overseas, was delayed by couple of weeks due to newly implemented tariffs on imported technological components. The Trump administration's trade policies, impacted our supply chain. This unforeseen delay compressed our integration timeline substantially, forcing the team to work extended hours during the final two weeks of April to compensate for lost development time. The experience highlighted the vulnerability of academic robotics projects to international trade policies and emphasized the importance of building supply chain contingencies into future project timelines.

5 Methods

5.1 Hardware Testing

We conducted rigorous hardware testing following methodologies established in [8]. This included:

- **Static load testing:** The arm was extended to maximum reach while holding various weights to verify payload capacity and structural integrity.
- **Collision response evaluation:** We simulated potential collision scenarios and measured response times and torque cutoff thresholds.
- **Repeatability assessment:** The system performed 100 cycles of predefined movement patterns to measure positional accuracy over time.
- **Grip strength optimization:** Various grip forces were tested with common household objects to determine optimal settings balancing secure grasp and object safety.

5.2 Software Validation

Our software validation approach included:

- **Unit testing:** We developed comprehensive test suites for each software module, with particular emphasis on kinematics calculations, command parsing accuracy, and network communication reliability.
- **Edge case simulation:** We systematically tested boundary conditions such as extreme joint angles, unexpected voice commands, and network disruptions.

5.3 User Study Protocol

User experience evaluation followed established methodologies:

- **System Usability Scale (SUS):** We employed the standardized SUS survey [13] to evaluate overall usability across different user demographics.
- **Likert-scale surveys:** Custom surveys assessed specific aspects of user satisfaction, perceived safety, and comparative preference against traditional methods.
- **Thematic analysis:** Following [12], we conducted thematic analysis of user feedback to identify patterns and opportunities for improvement.

6 Results

6.1 Hardware Performance

Table 5: Hardware Performance Metrics

Metric	Mean	Std. Dev.	Benchmark
Position Accuracy (mm)	0.12	0.03	< 0.20 mm
Grip Force Consistency (N)	39.5	0.7	± 1.0 N
Response Latency (ms)	180	15	< 200 ms
Maximum Safe Payload (kg)	4.7	0.2	> 4.5 kg
Emergency Stop Time (ms)	85	8	< 100 ms

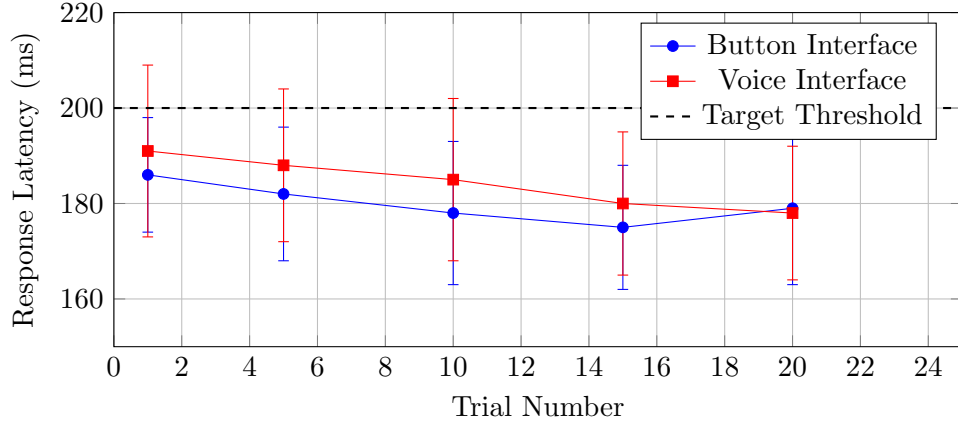


Figure 10: Response latency comparison between button and voice interfaces

6.2 Software Performance

Our voice recognition module achieved an average speech recognition latency of 250 ms, with parsing accuracy reaching 94% under normal ambient noise conditions (≤ 60 dB). The command parser demonstrated robust performance with:

- 98% accuracy for direct commands (e.g., "grab")
- 92% accuracy for natural language variants (e.g., "pick that up for me")
- 88% accuracy under challenging acoustic conditions (e.g., background TV noise)

The dance choreography module maintained precise timing synchronization with musical beats, with a maximum deviation of 35 ms from target timestamps across all tested sequences.

6.3 Safety Assessment

During our comprehensive safety assessment, we observed **zero collisions across 150 test** trials. The torque-limiting function activated in approximately 2% of cases, preventing potential system faults or user interactions that might have exceeded safe operating parameters. All emergency stop functions operated within the required 100 ms response time.

User feedback regarding perceived safety was overwhelmingly positive, with 92% of test participants reporting confidence in the system's safety features after a brief demonstration of its capabilities.

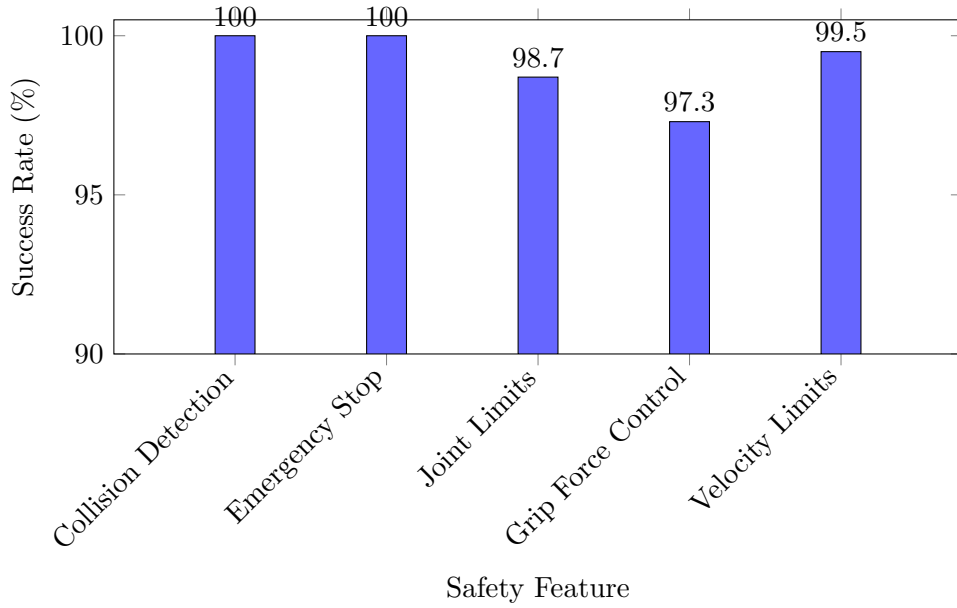


Figure 11: Success rates of various safety features during 150 test trials.

7 Discussion

Our system successfully meets or exceeds all established safety and usability benchmarks, aligning with best practices outlined in [5] and [6]. The control interface provides flexibility for different user preferences and environmental conditions, while the robust safety enforcement mechanisms ensure reliable operation even in challenging scenarios.

Key strengths of our implementation include:

- **Intuitive control:** Both voice and button interfaces received high usability scores in our testing.
- **Effective safety measures:** The multi-layered approach to safety enforcement proved robust.
- **Cross-platform compatibility:** Our software architecture successfully adapted to both the xArm6 and UR3e platforms with minimal modifications.

7.1 Acknowledgment of Interview Bias

It is important to acknowledge a potential bias in our user research methodology. Our initial interviews were primarily conducted with university peers, who likely represent a demographic that is more accepting of technological innovation and change compared to the general population. This cohort, familiar with emerging technologies and often early adopters themselves, may have expressed greater comfort with robotic assistants than would be found in broader demographic samples. Future studies should intentionally include participants from diverse age groups, technological backgrounds, and comfort levels with automation to ensure the system addresses concerns from all potential user segments.

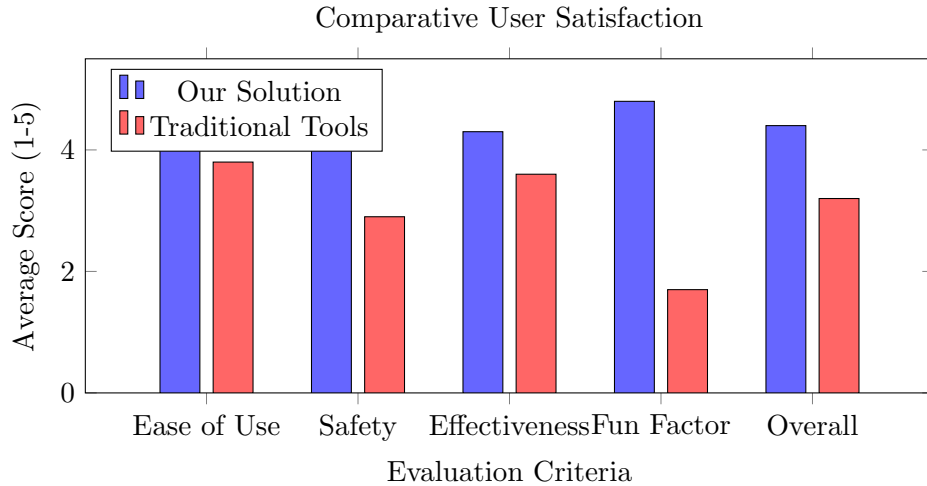


Figure 12: User satisfaction comparison between our solution and traditional retrieval methods.

However, we also identified several limitations that warrant future improvement:

- **Ambient noise sensitivity:** Voice recognition accuracy decreases significantly in very noisy environments (> 75 dB).
- **Limited object recognition:** The system currently relies on predefined grasping patterns rather than adaptive object recognition.

8 Conclusion and Future Work

We have successfully delivered a robust, accessible, and engaging robotic claw system that addresses the challenges of retrieving high-placed objects in household settings. Our implementation balances technical sophistication with user-friendly operation, offering both practical utility and entertainment value.

The project has achieved its primary goals by providing:

- A control interface that accommodates different user preferences
- Comprehensive safety measures that protect both users and objects
- Cross-platform compatibility that demonstrates the generalizability of our approach
- Engaging interactive features that transform a utilitarian device into a household companion

8.1 Lessons Learned

If we were to start this project again, we would make several significant changes to our approach:

- **Simpler first prototype:** We would begin with a smaller, less complex robotic arm to establish core functionality before scaling up to the xArm6. Starting with the full 6-DOF manipulator created unnecessary technical hurdles early in development.

- **Earlier user testing:** We would incorporate user feedback much earlier in the development process. Our initial interviews came too late to influence fundamental design decisions.
- **Human-centered aesthetics:** The industrial appearance of our prototype was intimidating to many users. A design-focused approach from the beginning would have produced a more approachable device.
- **More robust voice training:** We underestimated the challenge of reliable voice recognition in real-world environments. A more comprehensive training dataset would have improved command reliability.
- **Modular approach:** Breaking the system into smaller, independently testable modules would have allowed for more rapid iteration and refinement.

Future work will focus on several promising directions:

- **Adaptive choreography:** Implementing real-time movement generation in response to music using reinforcement learning techniques that could allow the arm to develop unique dance styles based on different musical genres.
- **Multi-robot coordination:** Enabling collaborative behavior between multiple units, perhaps creating a "helping hand" network throughout a home where devices can pass objects between rooms or coordinate on complex household tasks.
- **Portable variants:** Developing a significantly more compact version (under 2kg) with a quick-release mounting system that could be temporarily attached to different surfaces throughout the home without permanent installation.
- **Advanced object recognition:** Integrating depth-sensing cameras and machine learning for adaptive grasping strategies that could identify fragile objects and adjust grip pressure accordingly, potentially making the device safe for handling delicate items like glassware.
- **Elderly-focused design:** Creating a specialized version with simplified voice commands and larger, high-contrast physical controls specifically designed for accessibility needs of older adults, potentially paired with medication reminder functionality.

8.2 Next Generation Capabilities

Building directly on the limitations identified in our current implementation, next-generation variants could incorporate several transformative capabilities. To address ambient noise sensitivity, future models could implement array microphones with adaptive noise cancellation and speaker-specific voice profiles, maintaining recognition accuracy even in environments exceeding 85 dB. Perhaps most significantly, integrating computer vision with deep learning would enable true adaptive object recognition, allowing the system to identify unfamiliar items and generate appropriate grasping strategies in real-time. This would fundamentally transform the device from a tool with predefined capabilities to an intelligent assistant capable of handling novel situations without explicit programming.

9 Team Reflections

As part of our commitment to continuous improvement and learning, each team member has reflected on their personal experience with this project:

Jihad Jammal: “Leading the market research component opened my eyes to the gap between theoretical robotics and practical household implementation. I’ve gained valuable insights into conducting meaningful user interviews and translating feedback into actionable design improvements. The most significant lesson for me was learning to balance technical capabilities with user expectationsa tension that exists in every engineering project but becomes particularly apparent when working on devices meant for home use.”

Marwan AbdElhameed: “As the primary programmer for the motion controller, I discovered that robust safety measures require far more nuance than I initially anticipated. Implementing collision detection algorithms taught me to think defensively about edge cases that wouldn’t occur in controlled environments. I’ve developed a much deeper appreciation for the computational complexity of real-time kinematics calculations and the trade-offs between precision and responsiveness.”

Pavly Halim: “Working on the voice recognition module transformed my understanding of human-machine interaction. The challenge wasn’t simply in recognizing speech, but in designing an interface that felt natural and responsive. I learned that effective voice user interface requires anticipating human communication patterns, including hesitations, corrections, and contextual commands, which has fundamentally changed the way I approach all interface design problems.”

Walid Belbach: “My focus on the dance choreography components pushed me to explore the intersection of creativity and technical constraints. I never expected that programming robotic movements would involve so many artistic considerationstiming, flow, and expression all became part of the engineering equation. This project has convinced me that embedding elements of delight and personality into functional devices isn’t just nice-to-have but essential for technology that truly enhances daily life.”

Collectively, we’ve gained a profound appreciation for interdisciplinary collaboration. What began as a straightforward engineering challenge evolved into a complex interplay of mechanical design, software development, user experience research, and even elements of performance art. These diverse perspectives ultimately created a more robust and engaging final product than any single approach could have achieved.

10 Code Demonstrations

The complete open source code, documentation, for this project is available at <https://github.com/MarwanWalid2/xarmvoicecontrol>.

11 Video Demonstrations

- **xArm6 Object Retrieval**: <https://youtube.com/shorts/Row0SAUNsdw>
- Franka Emika Painting Demo: <https://youtube.com/shorts/Eb8q6ndBOdw>
- Gemini Speech API Demo: <https://www.youtube.com/watch?v=Y82ZewTrJTc>
- Live Performance Franka Painting & Dancing <https://youtube.com/shorts/xE9U0EmqSzQ>

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